



Public subsidies to private schools do make a difference for achievement in mathematics: Longitudinal evidence from Canada

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ABSTRACT

Selection into private schools is the principal cause of bias when estimating the effect of private schooling on academic achievement. By exploiting the generous public subsidizing of private high schools in the province of Québec, the second most populous province in Canada, we identify the causal impact of attendance in a private high school on achievement in mathematics. Because the supply of highly subsidized spaces is much higher at the high school level than at the grade school level, 94% of transitions from the public to the private sector occur at the end of primary school, we assume that these transitions are exogenous with respect to changes in transitory unobserved variables affecting math scores conditional on variables such as changes in income and child fixed effects. Using 7 cycles of data from Statistics Canada's National Longitudinal Survey on Children and Youth (NLSCY), we estimate the effect of attending a private high school on the percentile rank and a standardized math test score with different models (child fixed effects, random effects and a pooled OLS) and restricted samples to control for the degree of selection. The results, interpreted as a treatment on the treated effect show that the effect of changing schools, from a public grade school to a private high school, increases the percentile rank of the math score between 4 and 10 points and by between 0.12 and 0.36 of a standard deviation depending on the specification and samples.

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1. Introduction

Using their earlier research on the link between economic growth and cognitive skills Hanushek and Woessmann (2008, 2009), Hanushek and Woessmann (2010) in a recent Organisation for Economic Co-operation and Development (OECD) study conduct basic simulations/projections under different scenarios based on OECD countries and test score results (math, reading, science) from the Program for International Student Assessment (PISA) surveys of the 15-year-olds. The scenarios are for OECD countries: to increase average performance (cognitive skills) by 25 PISA points (0.25 standard deviation of improvement), bring each country to the average level of Finland (the country with the highest performance in PISA), and bring everyone up to a minimum skill level of 400 PISA points. The calculations assume that school reform begins in 2010, is completed within 20 years, while leading to a substantial increase in economic growth (roughly a one percentage point difference per year).

However, they point out that such objectives are not easily obtained: "These projections do not indicate how schools should be changed. Nor do they solve the political-economy issues of how any change can be achieved. They simply show the costs to inaction. Since the tests concentrate on the impact of schools, the evidence also suggests that school policy can, if effective in raising cognitive skills, be an important force in economic development. While other factors – cultural, health, and so forth – may affect

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the level of cognitive skills in an economy, schools also contribute to the relevant human capital. Economies with more human capital (measured by cognitive skills) innovate at a higher rate than those with less human capital, implying that nations with larger human capital in their workers witness more productivity gains.” (p. 10).

It is therefore not surprising that a tremendous amount of research on schooling choices and the impact of these choices on student outcomes have been done over the last 20 years. An empirical debate on the best policy to increase the level of achievement of students has become a perennial exercise for both policymakers and social scientists.

In research focusing on school choice and competition, several papers have raised many issues. First, do private schools (and charter, voucher and magnet schools considered by many as their proxies) improve student outcomes when compared to schools in the public system? Also, do these school effects, if any, persist over time (Horowitz & Spector, 2005)? Second, although some evidence suggests that a more competitive schooling market where parents have a larger set of choices can improve the academic performance of children (Hoxby, 2002), some critics argue that school choice may worsen outcomes of students left behind in public schools because of a cream-skimming effect, assuming that better peers have a positive impact on achievement (Dee & Fu, 2004; Walsh, 2009). Third, are schooling effects associated to families moving in better neighborhoods with desirable schools causal or due to sorting (Levine & Painter, 2008; Li, 2009; Liu, Mroz, & van der Klaauw, 2010).

This paper presents new evidence that attendance in non-religious subsidized private schools has a causal and positive impact on achievement in mathematics. Most empirical work on the impact of schooling choices made by parents for their children on academic achievement encounters the problem of spurious correlation of parental choice and achievement as unobserved variables such as parental tastes for the quality of education are correlated with both parental choice and achievement of children in school. We will argue that our approach, given the history and funding of the private schooling system in the province of Québec, the second most populated in Canada, solves this problem.

This paper addresses the issue of selection into private schools by exploiting the relatively generous public subsidizing of private high schools in the province of Québec to identify the causal impact of attendance in a private high school on achievement in mathematics without the use of instrumental variables. Because the supply of private high schools is much higher than for primary schools and that they are highly subsidized, 94% of transitions from the public to the private sector, in our sample, occur at the end of primary school. Nearly all students have to change school at the end of primary school and go to an unfamiliar secondary school and mix with unfamiliar students. The “switcher” bias in this case is minimal. We assume that this switch from public to private schools is exogenous conditional on variables such as income and child fixed effects. Using data from Statistics Canada’s National Longitudinal Survey on Children and Youth (NLSCY), we estimate the effect of private high school attendance on the percentile

rank and a standardized measure of a math test score with different models (child fixed effects, random effects and a pooled OLS) and restricted samples to control for the degree of selection. To the observed socio-economic covariates including a family functioning scale at different ages of the child as well as family income, we add in some regressions three behavioral scales to control for other potential confounders. The results show that attending a private school in Québec for those attending private schools in Québec (the treated) increases the percentile rank of the math score by between 4 and 10 ranks, depending on the specification and samples.

It remains possible that mechanisms such as Ashenfelter’s Dip (1978) could be in play. For example, a spurious positive effect of private school attendance will be estimated with a child fixed effects method if parents from higher income households, following a potentially temporary dip in school grades, react by sending their children to private school. A positive effect of private schooling would simply reflect a return of students to their normal levels of achievement, a type of mean reversion. Chay, McEwan, and Urquiola (2005) demonstrate this mechanism at an aggregate level. Using data from Chile, they show that the evaluation of public interventions in schools based on the test ranking of the schools will produce positively biased positive effects of interventions because they are done in response to a transitory poor score of schools who simply return to their normal levels after the intervention. Therefore, we must beware of temporary poor scores at an individual level when estimating the effect of private schooling. Given our identification strategy and some descriptive analysis presented later in the paper, we believe this issue is marginal in terms of its impact on the estimated effect of private schooling on achievement in mathematics.

The structure of the paper is the following. In Section 2, we provide a brief history of research in this area. In Section 3, we trace the long history of private schooling in the province of Québec and the uniqueness of the private school network among the Canadian provinces. Section 4 exposes the econometric methodology that is used to identify the causal effect (treatment on the treated effect) of private schooling on math achievement. In Sections 5 and 6, we detail the Canadian NLSCY data set used, and describe the explanatory variables in the model. We also present descriptive statistics as an introduction to Section 7 where the estimates of the effects of private school attendance are presented. Section 8 discusses the results as well as their external validity. The last section concludes with policy implications.

2. Recent research

There has recently been a diversity of initiatives in many countries to upgrade the performance of schools and student outcomes (such as independent schools in Sweden and the Netherlands, voucher systems as in Chile and the USA, and the charter movement in the USA), prompting research to consider with more attention the way schools are organized. The early issue was whether private schools make a difference compared to traditional public schools.

Seminal work by [Rouse \(1998\)](#), using the random variation from lotteries in the Milwaukee Parental Choice Program (MPCP), a publicly funded school voucher program and a child fixed effects estimation method, finds that non-sectarian private school attendance increases math scores by 0.08–0.12 of a standard deviation per year. This is comparable to some of the smaller estimates found in this paper. However, the Rouse sample does not include elite private schools while our sample includes children from all private schools. Recent evidence of an on-going longitudinal evaluation ([Witte et al., 2010](#)) suggests that students in the MPCP, which has grown over the past 19 years to enroll 19,803 students in 127 different private schools in 2008–2009, generally are experiencing achievement growth rates that are comparable to similar Milwaukee public schools students. Even students in the Milwaukee public school system have a variety of magnet, community, open enrollment, and inter-district school choice options available to them, so long as transportation funding holds out.

Using PISA results for countries where the number of students sampled and attending private schools is above a 10% threshold, and econometric estimation methods that deal with selection (IV, Heckman's two-stage method, propensity score matching), the results of [Vandenberghe and Robin \(2004\)](#) indicate no systematic benefits for private schools but some differentials in their favor, in particular for Belgium (where private schools are totally subsidized) and Brazil. For Chile, where a large voucher system operates for public and private schools, [McEwan \(2001\)](#) comparing six types of school for student achievement in Spanish and Math, finds a small advantage for Catholic voucher schools over most public and private schools before considering selection bias. The Chile experience with a package of resource-based interventions targeted at low-performing publicly funded schools has been criticized in some research. Among others, as we saw earlier, [Chay et al. \(2005\)](#) conclude that the empirical evidence on the positive impact of these interventions on test scores is largely an artifact of mean reversion.

In the United States, the not-publicly-subsidized catholic schools have for long been the main alternative for traditional public schools. Several influential studies have estimated the effect of Catholic secondary schooling on test scores, high school completion rates, and college attendance. [Neal \(1997, 1998\)](#) reviews the long history of research in this area, and [Altonji, Todd, and Taber \(2005a, 2005b\)](#) provide an overview of the discussion and references to the literature. These schools established across the United States are the largest and most homogeneous group of schools within the private sector and are found in a variety of geographical environments (urban, suburban and rural).¹ Results from multivariate regressions support the evidence that Catholic schools on average are

conducive to gains in attainment and achievement in comparison with the relevant public counterparts. However, one can argue that despite the numerous controls included in these analyses, private schools and their students are different by some unmeasured characteristics that lead to better schooling outcomes. These unmeasured traits may explain selection into private schools and the direct effects on the performance of students. Families who dedicate some of their resources to pay for private school attendance signal a preference for education. To minimize the positive bias of selection on the estimated effects of private schools, researchers have used a variety of exclusion restrictions such as religious affiliation, geographic proximity to Catholic schools, and the interaction between them to identify private school effects using IV methods. [Cohen-Zada \(2009\)](#) and [Altonji et al. \(2005a\)](#) evaluate the validity of instrumental variable strategies and conclude that none of the candidate instruments is a useful source of identification of the Catholic school effect, at least in currently available data sets. In another paper, [Altonji et al. \(2005b\)](#) develop new estimation methods using the degree of selection on the unobserved variables to assess the effect of attending a Catholic high school on a diversity of outcomes. They find no evidence of an effect on test scores, but conclude that private schools do have positive effects on high school completion rates and college attendance decisions.

In the USA, the charter school movement has gained momentum in the school choice reform debate. According to statistics presented in [Gleason, Clark, Tuttle, and Dwyer \(2010\)](#), at the end of 2009, more than 5000 charter schools served about 1.5 million students (approximately three percent of all public school students), in 40 states and the District of Columbia. Charter schools are intended to play a key role in school improvement under the existing Elementary and Secondary Education Act (No Child Left Behind) and the current federal reform support role for charter schools. Charter schools which operate under a contract (charter) with a government agency that subsidize them are provided a degree of autonomy from local school boards, and have fewer regulations with additional accountability requirements.

The effectiveness of charter schools in raising student achievement has become an intensely debated issue with a side debate on the best evaluation methodology and data set of charter schools to determine their impact. Most studies have focused on test scores (English and Math) because there are less data on other outcomes such as high school graduation and post-secondary school enrollment. The largest charter school study of its kind to date ([Gleason et al., 2010](#)) uses an experimental design comparing the results of winning and losing lottery students in middle schools (grades 5–8). Their results indicate that, on average, charter middle schools that hold lotteries are neither more nor less successful than traditional public schools in improving student achievement, behavior, and school progress; however, impacts varied widely across schools.

Findings from other studies (reviewed in [Gleason et al., 2010](#)) based on non-experimental methods have been mixed. The studies that cover a wide span of states and/or districts, found no impacts in reading and mathematics. [Zimmer, Gill, and Booker \(2009\)](#) found that impacts on test

¹ According to the Annual Data Report of the National Catholic Association, the United States had 7498 Catholic schools in 2006–2007, including 6288 elementary schools and 1210 secondary schools. In total there were 2,320,651 students, including 1,682,412 students in elementary or middle schools and 638,239 in high schools.

scores were negative or not significant, but favorable high school graduation and college enrollment results suggest the possibility of long-term benefits. CREDO (2009) examined charter schools in 14 states and found average impacts to be negative and significant.

The non-experimental approaches have been criticized (Hoxby, 2009; Hoxby, Murarka, & Kang, 2009). The experimental studies focusing on the charter school sites in large urban areas found that charter schools serving large populations of disadvantaged and racial/ethnic minority group students had positive impacts (Hoxby et al., 2009, for New York charter schools). On the other hand, CREDO (2010), for their New York charter schools evaluation, using the same matching methods as for the pooled national study (CREDO, 2009), also find that the effects of charter schooling on student academic performance are positive. However, a more critical view is expressed by Gleason et al.:

“First, while the experimental design minimizes selection bias, and thus ensures a high level of internal validity, the study’s external validity is limited. Second, an assumption underlying the study’s experimental design is that study charter schools did not influence the performance of the traditional public schools and other schools against which the charter schools compete for students. If the study schools influenced these schools, then both the treatment and control group students in the study sample could have been influenced by the study charter schools, potentially affecting the study’s impact estimates.” (Gleason et al., 2010)

To summarize, the evidence on the effects of private schools or charter schools is mixed, although, when effects are found, they are generally positive. The next section provides a brief history of private schools in Québec and lays out the foundation of our identification strategy.

3. Québec’s private schools

This section describes the changes of Québec’s education system over time and highlights its distinctive features compared to the education system of the other Canadian provinces. We present the sources for the revenues of private schools in Québec, the evolution over time of the number of students enrolled in public and private schools, and, finally, the measured gaps in test scores between the students of these schools.

Among all the ten Canadian provinces, Québec has the most extensive sector of private schools. Table 1 shows the number of students in public and private primary and secondary schools for the school year 2003–2004 in Québec and in the four other Canadian provinces which dedicate public funds to private schools as well as total subsidies. In the other five provinces (including Ontario, the most populous with 39% of the population) there are also private schools, but they are not subsidized and enrolment is very low (see also Table 8).

In Québec, the tradition of subsidizing private schools has a long history. From the 1950s to the end of the 1960s, most subsidized schools were “classical elite colleges” (the study of Latin or Greek or both was mandatory as well as

Philosophy). The curriculum lasted eight years following the completion of primary schooling after which graduates had access to university education and, in particular, to professional faculties such as medicine, law, dentistry, optometry, pharmacy, etc. Otherwise, graduates from public schools, after 12 years of schooling, had access to a more limited number of university programs. At the end of the 1960s, the Québec government substantially modernized the education system. Public kindergarten was created; the number of grades in primary school was reduced from 7 to 6 years; the number of grades in secondary school was also changed from 6 to 5 years; most “classical private colleges” became public colleges (and new ones were created over the years).² Québec’s colleges form a distinct schooling level compared to the schooling system in the other provinces.

With the changes brought to Québec’s education system, the public financing of private schools was also revised in the early 1980s. Two important rules were then adopted and are still in place today.

Firstly, and abstracting from a small “capital subsidy” for installations, the subsidy for operating costs³ per student given to private schools is revised each year according to the subsidy per student given to public schools by level of education (kindergarten, primary, secondary, college). The yearly raise in subsidies to private schools reflects the percentage increase of subsidies to the public system schools (for employee wages and other costs).

Secondly, the percentage of operating costs covered by the subsidy was lowered from 80% to approximately 60% for all schooling levels. The subsidy per student depends on the schooling level and is the same for all private schools. Table 2 presents, for the school year 2004–2005, the subsidies per student for a private institution at the kindergarten, primary, secondary school and college levels; they are respectively, \$3006, \$2582, \$3556 and \$4757, to which the government adds respectively \$93, \$93, \$139 and \$514 per student for installations. Hence, the subsidy per student in private secondary schools is considerably larger than for private primary schools.

Evidently, the sources of revenue differ between Québec’s public and private schools. They are presented in Table 3 for public and private schools and school year 2004–2005. For private schools, almost half of the operating expenses are covered by subsidies and a third is generated by registration, admission, and school fees. Moreover, the government requires that the fees of subsidized private schools not exceed the subsidy per student. Table 4 shows the authorized maximum fee according to regulations and the actual average fee charged by private schools by schooling level. Very few private schools fix their fees to the maximum permitted. Administrative statistics

² Many of these “classical” schools were run by a catholic order (brothers, sisters, priests) although most of the teachers were secular. The Catholic Church was a very important player in Québec society until 1960. The best private high schools were almost all catholic and a large number operated throughout the province. Therefore, because of historical reasons, the supply of private high schools is relatively high in Québec.

³ The operating costs consist mainly of teachers and non-teaching salaries and other costs except the location value of premises.

Table 1

Number and proportion of primary and secondary level students in public and private schools in the Canadian provinces that subsidize private schools, and total subsidies, 2003–2004.

Province	Number in public schools	Number in private (subsidized) schools	Percentage in private schools (subsidized)	Subsidy per student for eligible private school	Total subsidies million dollars
Québec	989,838	115,200 (106,681)	10.4 (9.63)	Kindergarten: \$2976 Primary: \$2578 Secondary: \$3555 Average: around 60%	397
Manitoba	188,498	14,290 (12,861)	6.51 (5.89)	Kindergarten: \$1690 Primary: \$3381 Secondary: \$3381 Average: around 50%	45
Saskatchewan	174,263	2627 (1655)	1.36 (0.86)	Primary: \$0 Grade 9: \$4723 Grades 10–12: \$5045–5271	8.3
Alberta	552,592	28,500 (27,209)	4.81 (4.59)	Kindergarten: \$2227; 100% Primary: \$2672; 60% Grades 7–9: \$2672; 60% Grade 10: \$3058; 60% Grades 11–12: \$4580; 60%	75–80
British Columbia	614,671	63,321 (54,888)	9.27 (8.04)	Category 1: \$3001; 50%; 72% of students in private sector Category 2: \$2088; 35%; 15% of students in private sector Weighted average: \$2837	156

Marois (2005).

Table 2

Public subsidy per student to Québec's private schools by schooling level, selected years, in Canadian dollars.

Year	Kindergarten schools	Primary schools	Secondary schools	College
1997–1998	2275 (+82)	2092 (+82)	2919 (+122)	NA
1998–1999	2297 (+82)	2108 (+82)	2944 (+122)	NA
2000–2001	2496 (+85)	2292 (+85)	3179 (+122)	NA
2002–2003	2807 (+88)	2421 (+88)	3331 (+131)	NA
2004–2005	3006 (+93)	2582 (+93)	3556 (+139)	5271 (+514)
2005–2006	3064 (+95)	2808 (+95)	3612 (+142)	NA

Ministry of Education Leisure and Sports (MELS), Budgetary Rules for Schools under Agreement, annual.

Note: The subsidy is based on teaching and non-teaching personnel, other costs (plus location value of premises in parenthesis).

Table 3

Sources of revenue for Québec's public and private subsidized schools in percentage, by schooling level, 2004–2005.

Revenues	Primary and secondary schools	Colleges
Public sector		
Subsidies	77	86
School taxes	14	–
Admission, registration and schooling fees	–	3
Other revenues (gifts, profits from ancillary activities, etc.)	9	11
Private subsidized sector		
Subsidies (including administrative agreements)	48	55
Admission, registration and schooling fees	28	30
Other revenues (gifts, profits from ancillary activities, etc.)	24	15

Ministry of Education, Leisure and Sports (MELS), Statistics of Education, 2005, p. 231.

(not shown here) for 169 subsidized private schools indicate that, on average, the fee is 68.3% of the authorized maximum fee and that this gap varies by region (from 39% to 84.3% in Montreal, the largest city in the province of Québec), which suggests that families are price sensitive and that the market is competitive.

All private schools must have a permit delivered by Québec's Ministry of Education to operate. The Ministry treats the demands for the permit as well as renewals, and modifications to the permit which specifies the maximum number of students on the school's premises. To be accredited and to receive subsidies, the curriculum must be approved by the Ministry. All students in grades 4 and 5 of secondary private schools must also take the standard French, English, History, Math, and Physics exams of the Ministry, as do the students of public schools. These are exit exams that students must pass to obtain the Ministry high school diploma.

Almost all private schools are subsidized (see Table 1). They represent approximately 90% of the primary and secondary private schools. Two “types” of private schools are not subsidized. The first type consists of a very few “elite”

Table 4

Authorized maximum and average schooling fees in Québec's private schools, by education level, and school years 2003–2004 and 2004–2005.

Level	Year	Maximum fee according to regulation	Average fee asked	Number of schools with maximum fee
Kindergarten	2003–2004	\$2886	\$1650	1 out of 48
	2004–2005	\$2924	\$1679	
Primary	2003–2004	\$2488	\$1724	2 out of 72
	2004–2005	\$2518	\$1813	
Secondary	2003–2004	\$3421	\$2122	2 out of 142
	2004–2005	\$3464	\$2219	

Ministry of Education, Leisure and Sports (MELS), Financial Reports of Subsidized Private Schools, and Budgetary Rules for Schools under Agreement, annual.

Table 5

Student enrollment in Québec's public and private schools, by schooling level, and by school year, 1994–1995 to 2007–2008.

School year	Public schools					Private schools		
	Kindergarten	Primary	Secondary	College	Kindergarten	Primary	Secondary	College
1994–1995	86,091	522,714	421,467	163,990	3821	24,681	76,839	16,979
1995–1996	91,307	520,882	415,491	160,405	4053	24,595	75,749	16,608
1996–1997	91,652	525,973	410,538	161,684	4167	24,847	74,885	16,528
1997–1998	91,001	531,816	404,333	158,032	3098	25,350	73,806	16,348
1998–1999	87,441	538,394	395,246	157,209	3786	26,155	72,837	15,064
1999–2000	85,053	544,452	382,648	150,055	3918	26,941	72,635	12,074 ^a
2000–2001	83,073	546,444	373,504	150,845	4010	27,831	73,343	11,805 ^a
2001–2002	80,006	543,546	370,197	144,111	4362	28,995	74,964	11,606 ^a
2002–2003	76,421	533,276	376,409	148,726	4303	29,462	77,913	11,097 ^a
2003–2004	72,223	517,996	385,139	147,817	4372	29,473	81,310	10,381 ^a
2004–2005	70,185	498,369	395,120	147,170	4421	29,917	84,066	10,924 ^a
2005–2006	69,344	478,772	401,407	148,093	4625	30,461	86,561	10,834 ^a
2006–2007	69,043	460,502	402,946	146,636	4776	31,101	88,203	10,953 ^a
2007–2008	68,906	446,141	394,673	151,658	4832	31,533	88,694	12,045 ^a

Ministry of Education, Leisure and Sports (MELS), *Statistics of Education*, annual.

Note: Regular education, full-time and part-time students.

^a Subsidized spaces in colleges.

primary and secondary schools who offer enriched curricula and high levels of pedagogical supervision as well as other activities. All are in the metropolitan region of Montreal. Their schooling fees are much higher than the maximum permitted to be a subsidized school. The second type, the majority of unsubsidized private schools, consists of “commercial” schools or colleges which offer very specific lower-skill training (hairdressing, secretarial skills, paramedical assistance, travel agent, etc.).⁴ In school year 2003–2004, private schools are distributed as follows by education level: 130 at the primary level, 140 at the secondary level, 59 at both the primary and secondary levels, 24 at the college level (16 offer both the secondary and college levels). Hence, the majority of private schools are found at the secondary level of education (215 schools). By comparison, there are respectively 1754 primary, 428 secondary, and 198 combined primary with secondary public schools (52 public colleges, and all the 19 universities are public). Therefore, there are 26 more private secondary level schools than private primary schools, and secondary level schools are much larger, thus explaining the much larger supply of private high school places compared to private primary school. For school year 2007–2008, there were 32,000 students enrolled in private primary schools and

89,000 students in private secondary schools; for public schools the enrolments were respectively 395,000 (secondary level) and 446,000 (primary level).

Table 5 presents the number of students enrolled in Québec's public and private schools, by education level and for the school years 1994–1995 to 2007–2008. The statistics show a declining trend of total enrollment in public schools at the primary and secondary levels. This trend is similar in Canada and can be explained by the declining rates of fertility over the years. In contrast, there is an upward trend in the private sector at the primary and secondary levels, but not at the college level. The private enrollment ratio varies by province as can be seen from column 3 in Table 1 (and in Table 6 constructed from our data set). One Québec student out of ten is enrolled in the private sector. Only British Columbia has a similar rate. A rapid calculation from Montreal's School Boards statistics indicates that more than 20% of students are enrolled in private secondary schools. For the purpose of this study, it is important to stress the fact that few Québec families enroll their child in a private primary school. Statistics for the last year available from the survey data used in our study are presented in Table 6. They show for the school year 2006–2007, only 6.15% of Québec's students are in a private school at the primary level. On the other hand, almost one out of five (18.46%) students attends a private school at the secondary level (Table 6, last column). In the other provinces the proportion of students in private schools at both levels is very

⁴ Most of the technical skill training is also offered in public schools which all have training centers for manual trade occupations.

Table 6

Weighted number of primary and secondary school students in the public and private sectors by province and cycle of the NLSCY.

Provinces	Type of schools ^a	Cycle 1: 1994–1995	Cycle 2: 1996–1997	Cycle 3: 1998–1999	Cycle 4: 2000–2001	Cycle 5: 2002–2003	Cycle 6: 2004–2005 ^b	Cycle 7 2006–2007 ^c
QC	Public	389,035	540,221	665,036	685,545	631,457	536,267	556,935
	Private	21,441	45,413	68,833	72,682	77,753	81,057	83,847
	%Private	5.22	7.75	9.38	9.59	10.96	15.25	13.09
MAN, SASK, ALB, BC	Primary							18,790
	%							6.15
	Secondary							65,571
	%							18.46
	Public	548,162	789,440	970,069	924,861	835,658	676,955	815,798
	Private	26,034	48,251	61,347	51,677	50,684	44,814	54,697
NFL, PEI, NS, NB, ON	%Private	4.53	5.76	5.95	5.29	5.72	6.65	6.28
	Primary							29,827
	%							6.99
	Secondary							25,507
	%							5.69
	Public	832,403	1,191,455	1,419,990	1,410,145	1,287,780	985,282	1,294,122
Total	Private	32,728	46,804	36,498	43,737	41,332	34,048	59,862
	%Private	3.78	3.78	2.25	3.00	3.10	3.34	4.42
	Primary							32,958
	%							4.81
	Secondary							27,074
	%							4.02
Total	Public	1,769,600	2,521,116	3,055,095	3,020,551	2,754,895	2,238,876	2,865,261
	Private	80,203	140,468	166,678	168,096	169,769	159,880	198,406
	%Private	4.53	5.57	5.46	5.57	6.16	7.14	6.92
	Primary							81,575
	%							5.76
	Secondary							118,152
	%							8.00

Authors' calculation from micro-data Cycles 1–7 of the NLSCY. QC: Québec; MAN: Manitoba; SASK: Saskatchewan; ALB: Alberta; BC: British-Columbia; NFL: Newfoundland; PEI: Prince-Edward-Island; NS: Nova-Scotia; NB: New-Brunswick; ON: Ontario. Students are weighted by their longitudinal weight in each cycle.

^a The question on type of school is asked only for children in school and aged 15 years or less. At Cycle 1, all children are aged 11 years or less, and almost all are in grade (primary) school.

^b There are no children aged 6 to 9 (in grade school) at Cycle 6 of the survey which explains the drop in the number of students.

^c At Cycle 7 of the survey there are no children aged 10–11 years.

similar (Table 6, last column) which suggest that parents choose to enroll their child from grade 1.

3.1. Outcomes in public and private schools

School performance can be measured by high school graduation rates, the average number of years used to obtain a high school degree, and access to postsecondary education (college and university). Along all these dimensions, private schools in Québec do better than public schools.

Québec's Ministry of Education administers unique exams in French and English, History, Math, and Physics, for students in grades 4 and 5 in all secondary level schools, whether they are public or private. As of June 2005 and for all schools, the overall average score and pass rate in these exams were, for public and private schools respectively, 73.3% and 83.3%. In private schools, the overall average score was 80.8%, compared to 71.4% for public schools. The passing rate was 80.6% for public schools and 94.5% for the private schools. For the 2002 cohort, 53% of students in public schools obtained their high school Diploma after 5 years (the "normal" length after primary school)⁵ compared to 83% for students in private schools.

A contentious issue in the public debate about educational choices is the role of selection, based on academic performance, into private schools. Opponents of the use of public funds to finance private schools, especially the Association of Public School Boards, argue that the higher performances of the private sector are the "natural" result of the selection of "high ability" students. If an important fraction of high achievement students enroll in private schools, there is a decline in both the overall quality of public school students and of the performance of the lower ability lesser-skilled students remaining in public schools (the "cream skimming effects"). This assumes implicitly or explicitly that positive peer effects exist, that sorting by ability or socio-economic status would hurt low-achieving students (considered also as of low-income status). Finally, opponents to the subsidization of private schools argue that an objective comparison of the academic performances of public and private schools requires comparing students from the same socio-economic backgrounds, which is what we attempt to achieve in this paper.

Private schools can and do select their students.⁶ In a 2006 document on admissions, the Federation of Private Schools and Institutions (FEPP, 2006) reports that "70.0% of students who took an admission exam for grade one of secondary school were admitted, 17.6% had their application rejected because of space limitations, and 5.4% were

turned away because the school did not have the specialized human resources to respond to the special needs of these students" (p. 3). This suggests that almost 25% of the candidates have seen their applications in private schools rejected because of their results at the entrance exam. Results from a 2005 survey by the Federation (2006) indicate that many institutions have a waiting list for admission to their secondary level schools. Selection is practiced when admissions are higher than available spaces. From the survey (124 respondents of the 150 institutional members), half declare that they accept all students and the other half report that they must reject applications because of space considerations.

3.2. Competitive responses of the public schools to private school innovation

In the mid-nineties, the Ministry of Education decided to make public the overall provincial results of the provincial common exams and pass rates for each school. Over the years, some institutes ([The Fraser Institute](#); [The Montreal Economic Institute](#)) use the student results in core academic courses to calculate an overall rating of each school's academic performance. The annual ranking was and is also published by a popular bi-monthly magazine. The published ranking showed (and still shows) most of the private secondary schools on top of the list and it has become eagerly awaited by parents shopping for secondary schools in the fall.

During that same period, School Boards who manage the public schools reacted more vigorously to the flight of students to private schools by offering more selective and particular pedagogical projects to retain more skillful students, and promote more attractive programs from the point of view of parents and students. The number of public schools where the school is totally dedicated to a particular project remains small: sixteen in all; four with an international program; five with an "enriched" program; 1 with a foreign language program; 3 with an arts program; 2 "alternative" schools and 1 school for the re-integration of high school dropouts. They draw students from across the normal boundaries defined by School Board authorities.⁷ On the other hand, more schools have developed a within school particular project. For example, according to administrative data for school year 2005–2006, 93 public high schools have adopted a limited international program. Table 7 presents the number of students in primary and secondary public schools enrolled in particular pedagogical projects by type. Unfortunately, our data set cannot identify public school students who are in "special" programs. Because the creation of these programs rose with the increase in private school attendance, we consider these reactions of the public schools as a competitive response to the quality and diversity of programs offered in private schools.

⁵ The age eligibility for kindergarten is 5 on September 30th. High school graduation age should be at age 16 or 17. School attendance is mandatory until the students have reached the age of 16.

⁶ The selection process is not very rigorous at the primary school level. The schools who offer both the primary and secondary levels generally admit to the secondary level a student that is enrolled at the primary level if she succeeds at this level. The tradeoff for families is therefore to strongly increase the probability of admission at the secondary level at the expense of paying admission fees for the 6 years in private primary school.

⁷ In Canada and Québec like in the USA, public school choices are constrained by residential location.

Table 7

Number of students in primary and secondary public schools enrolled in particular pedagogical projects by type, 2004–2005.

Type of program	Number of students
Regular program with arts (designated or concentration)	12,882
Regular program with sports (designated or concentration)	13,217
International program	24,317
Regular program with computer science	2681
Regular program with foreign languages	14,370
Regular program with sciences	2518
Regular program with a diversified project	4244
Alternative (non traditional) school	3091
Total	77,320

Higher Council of Education (2007) from the Ministry of Education, Leisure and Sports file on students' enrolment.

4. Analytical framework

Endogeneity is the main econometric problem encountered when trying to measure the effect of private schools on the academic performance of children. Nonrandom selection into private schools is an issue most studies on the subject try to tackle, in the absence of experimental data. Unobservable background characteristics (correlated with achievement) of most students can lead parents to enroll their child into a private school. Moreover, characteristics of private and public schools are likely to influence their effectiveness.

Obviously, OLS regressions will produce a positively biased effect of private schooling on math scores or ranking, children from wealthier families being drawn to private schools. Even controlling for income, OLS estimates are biased. Children in private schools live with parents who on average place a higher value on education and purchase inputs helpful for development, both unobserved by the econometrician. We will use a child fixed effects regression method to compute consistent estimates of this effect.

Our identification strategy is simple. It is based on the idea that moving from a public grade school to a private high school in the Province of Québec is uncorrelated with changes in unobserved variables affecting achievement in mathematics during that same time period. We believe that the principal cause that explains the relatively large (compared to other jurisdictions across Canada) transitions between public primary school and private high school is the large increase in the supply of rather inexpensive private school places when the child transits to the high school level in Québec. Therefore, we rule out spurious effects caused by mechanisms such as Ashenfelter's Dip that would bias the estimated effects of private schooling using panel data and child level fixed effects. Without totally dismissing that such a mechanism is playing out in areas with high fees for attending private schools, we surmise that it plays a very small role in the case of a sample of children who are moving in a private school in Québec as they leave grade school in the sixth grade.

As we stressed in the last section, the subsidy per student of Québec's private high schools is more important than that of its private elementary schools and the sup-

Table 8

Number of transitions and percentage of total transitions from public to private schools, Québec, and Cycles 1–7 of the NLSCY.

Transitions by schooling level ^a	Number	% of total
In primary school ^b	6	3
In secondary school ^c	6	3
Between primary and secondary	189	94
Total	201	100

Authors' calculation from the NLSCY.

^a Excludes the transitions from private to public schools (30 in all).

^b Primary school is defined as 3rd to 6th grade.

^c Secondary school is defined as 7th to 10th grade.

ply of places is much higher. Moreover, this variation in the subsidy or supply of places does not depend on any unobservable characteristics of children. It is thus equivalent to an exogenous change in private schools spaces and prices. As families are price and supply sensitive, this variation explains why the percentage of students enrolled in private schools is more important in high schools than at the elementary level. Hence, measurements of the effect of Québec's private schools on the academic performance of Québec's children should not suffer from simultaneity bias with a child fixed effects estimation method.

Table 8 shows that a very large proportion of Québec children in our sample (described in the next section) move to the private schooling sector after grade school in reaction to the large number of private schools available at the high school level as shown in the preceding section. Out of 201 children transiting from public to private school in our sample, 94% are observed transiting from primary public school to secondary private school. We assume that a sample of children composed of children observed in primary school in one cycle of the NLSCY and in high school the next will be less sensitive to Ashenfelter's Dip and will provide us with credible causal estimates of the effect of attending private school on achievement in mathematics. We rule out instrumental variables estimation using the regional variation in subsidies because it is fixed across regions. Also, the time-series variation in subsidies is almost non-existent and we have only seven years of data.

Our modeling strategy to capture the effect of private school attendance on academic performance is progressive and simple. First, we select all of the NLSCY's children aged 7 to 15 years with at least two math scores.⁸ This measure of academic performance is regressed on socio-economic covariates including a private school dummy variable and child fixed effects. Practically, we estimate the parameters of the following child fixed effects model using the panel features of the NLSCY:

$$T_{ia} = \alpha \times \text{private} + X_{ia}\beta + \mu_i + e_{ia} \quad (1)$$

where T_{ia} is child i 's percentile math score at age a ; *private* a dummy variable equal to one if the child attends a private school; X_{ia} the observed socio-economic covariates at age a ,

⁸ A child may not be observed with consecutive math scores, i.e. a child may drop out of the survey for one wave and then reappear. Statistics Canada has eliminated from the survey a few family respondents missing two consecutive cycles.

including household income and a constant; μ_i is the child's fixed effect; and e_{ia} the residual. The vector X_{ia} may be “augmented” to include three behavioral scales that measure a child's hyperactivity (hyperactive–inattentive), anxiety (anxiety and emotional disorder), and aggression (conduct disorders–physical aggression). Including these behavioral scales is an additional means to control for a possible endogeneity problem that might bias our estimate of the effect of private school attendance on children's academic performance.

After estimating the model with all children who have at least 2 math scores, we restrict our sample by considering only children at the end of primary school (5th and 6th grades) and at the beginning of secondary school (7th and 8th grades). The first sample should be more prone to bias due to Ashenfelter's Dip because children may change from a public to a private school after experiencing a transitory bad period in school. The second sample provides a variation in the private schooling variable that is exogenous to temporary variations in unobserved factors related to achievement levels in school.

In addition to the child fixed effects model (1), we also estimate a random effects model and a pooled OLS model to observe the bias present in methods that do not provide a control for selection. Because of the probable heterogeneity of the effect of attending a private school on achievement, and that there are almost no transitions from private to public schools, we interpret our estimated effect of private schooling as an average treatment on the treated effect in a difference-in-differences model, that is the effect of private school attendance on children who attend private schools (see Blundell & Costa-Dias, 2009).

5. Data set

The data used for our empirical analysis are provided by Statistics Canada's National Longitudinal Survey of Children and Youth (NLSCY) which is a probability survey designed to provide information about children and youth in Canada. The survey covers a comprehensive range of topics including childcare, schooling, physical development, learning and behavior as well as data on the social environment (family, friends, schools and community). The NLSCY began in 1994–1995 and data collection occurs *biennially*. The unit of analysis for the NLSCY is the child or youth. Information concerning each selected child between 0 and 17 years of age and the family are provided by the person most knowledgeable (PMK) of the child. The PMK is usually the child's mother (more than 90% of the time), but it could also be the father, a step-parent or an adoptive parent who lived in the same dwelling.

A sample of 22,831 children, aged 0–11 years was selected at Cycle 1 (1994–1995) of the survey, in each of the 10 provinces, making up the longitudinal sample. To reduce the response burden on families with several eligible children, the number of children selected was limited to two per family at Cycle 2 (1996–1997). Therefore, some children were dropped from the original sample (16,903 children remained in the longitudinal sample). The rule changed to one child per household without exception at Cycle 5 (2002–2003). Because of the design of the survey,

no separate longitudinal data fixed effects sibling analysis can be conducted. The last data set available is Cycle 7 (2006–2007), where the children are aged from 12 to 23.

The test is administered to children who were at least enrolled in their second grade in primary school and aged from 7 to 15.⁹ Therefore, not all children from the original longitudinal sample (1994–1995) can be used for this paper's estimations. We use data on children aged 7–15 with at least two test scores.¹⁰

5.1. Math scores

The CAT/2 test is a shorter version of the *Mathematics Computation Test* taken from the *Canadian Achievement Tests*, 2nd edition. The CAT/2 is designed to measure basic skills in math (addition, subtraction, multiplication and division on integers, etc.). The master files present both the CAT/2-raw and -standardized scores. The raw score is simply the number of correct answers to the test. The difficulty of the test varies with the schooling grade of the child. Thus, there are different tests depending on the school level the child is in. Statistics Canada standardizes the raw scores using a sample of Canadian children from the ten provinces called “the normative sample”. This sample was chosen by the Canadian Testing Center. The standardized scores are obtained using sub-samples (by schooling-grade) of the normative sample.

A significant proportion (approximately 35%) of children in two school grades obtained a perfect score when they passed the tests in 1994–1995 (at Cycle 1 of the survey). Therefore, subsequent cycles added more versions of the tests. These “newer” versions of the tests were still based on the school level of children but with a clearer distinction made between school years. For example, in 1994–1995, children in their second and third school years (who are 7 and 8 years old) took the level 2 test. Two years later, children of the same ages took two different tests: the level 2 test for children in their second school year (7 years old) and the level 3 test for children in their third school year (8 years old). Hence it is possible that children of different ages passed the same test in a particular cycle. Preliminary results with the standardized scores (presented in Table A3) showed strong negative period effects. Given Statistics Canada's interpretation of the scores, this translates into a serious deterioration in math achievement across Canada. Because this made little sense, considering that the mean math performance of the 15-year-olds Canadians in successive OECD PISA' surveys (2004, 2007) has not declined, the Hazen method¹¹ was applied to calculate another dependent variable for math achievement,

⁹ All the 16- and 17-year-olds take only once a different math test (reading comprehension, problem-solving and decision making).

¹⁰ At Cycles 2 and 3, there was both a math and a reading test. At Cycle 4, only the math test was administered. The reading test was removed because of time constraints. It was decided that only one test could be administered. The math test was chosen as it had been administered in all previous cycles.

¹¹ The formula for the Hazen percentile (scores are reported as percentile ranks) is given by $(i - 0.5)/n$, where i is the rank of the individual and n the number of individuals in the group for which the percentile is computed. <http://www.stata.com/support/faqs/stat/pcrank.html>.

percentile ranks based on the standardized scores. The percentile ranks are always calculated with regard to the level of the test the child passed and for each cycle of the NLSCY used. We also use as an alternative dependent variable, because the percentile rank variable is truncated, the math score in units of a standard deviation (standardized to have a mean 0 and a standard deviation 1).

We have fewer observations for Cycles 1–3 compared to Cycles 4–7. From Cycles 1–3, the test was administered by the child's teacher once the PMK of the child and the School Board had given consent, and the math teacher available. At Cycle 1, the NLSCY produced results for approximately 50% of eligible children; at Cycle 2, the percentage was 74% and at Cycle 3, it was 54%. From Cycle 4 of the NLSCY, in order to minimize the time dedicated by teachers to the survey and to avoid disturbing class activities at the end of the school year, the math test is administered at the home of the child rather than in school, and almost all eligible children are observed with a test score. This explains partly why the sample is larger from Cycle 4.

6. Control variables and descriptive statistics

Tables A1 and A2 present the means of the socio-economic control variables (characteristics of the children, PMK, and families) respectively for the overall sample and the constrained sample where the children are restricted to be in a cycle when they are in primary school and in the next cycle are in secondary school. For child characteristics, we control for: age in months, gender, if the child has repeated his/her last grade, has a chronic condition, changed school (other than a transition from public to private school), age and number of siblings, and scores on three behavioral scales constructed by Statistics Canada and mentioned above: hyperactivity, anxiety, and aggression. For children aged 7–9, the behavioral scores are reported by the PMK of the child. For children aged 10–15, the scores are self-reported. For the PMK, we control for: mother's age at the child's birth, level of education, immigrant status, family type, size of community and family income (2002 dollars). A family functioning score is also used as a control.¹² This scale provides a global assessment of family functioning and an indicator of the quality of the relationship between family members. The total score varies between 0 and 36, a high score indicating family dysfunction.

7. Econometric results

We start our analysis with the results obtained with the percentile rank of the child for the math test as dependent

variable and presented in Table 9. We show the pooled OLS, random effects and fixed effects estimates of the effect of private school attendance on rank and with two specifications (without and with behavioral scores added to the socio-economic control variables). For Sample 1 (first panel) which includes all public and private school children, all estimates of the private school effect are statistically significant with the OLS estimate being the largest (9.87), as expected given that unobserved factors affecting private school choice are positively correlated with achievement (e.g. parental involvement in education). The child fixed effects estimate is considerably lower but shows that private schooling can make a difference for children with parents who choose private schooling as we find that the effects are 5.39 and 5.24 percentile ranks. The random effects estimates are between OLS and fixed effects showing as the fixed effects estimates that individual heterogeneity plays an important part in math scores, which is not a surprising result.

Our second set of estimates (Sample 2, second panel of Table 9) is obtained with children who are observed in either the fifth grade or sixth grade at the primary level and then in the next wave in high school. Unfortunately, there is a gap of one grade because the children of the NLSCY are observed every two years. This sample should be the least affected by a mechanism such as Ashenfelter's Dip. Here, OLS estimates and random effects estimates are very similar (10.3 versus 8.65). This occurs because very few children are in private school at the primary level. Hence, almost all the variation in the private schooling variable occurs at the secondary level. Adding the behavioral scores of the children has a very marginal impact on the estimated parameters of the private school effect. Hyperactivity and aggression are generally statistically significant variables, having a negative effect on the math scores. Whereas, anxiety has a positive impact of the percentile rank but is not significant for all cases.

The fixed effects estimates are larger (7.28 and 6.61) for this sample compared to the first. This could be explained by the fact that when transitions are done before the change from primary to secondary or during the secondary level, it could be because students are doing badly and parents react by sending them to private schools. This type of student could react less strongly to the benefits of private schools.

Because there is a considerable number of missing math scores in the first three cycles, we performed the same regressions with children with observed scores in only Cycles 4–7 so that bias due to attrition is considerably reduced as response rates for the math test are close to 100 percent for these four cycles. The results appear in the panels for Samples 3 and 4. Despite the loss of a considerable number of observations, we continue to observe statistically significant effects of private schooling in all samples and specifications except one. The fixed effects method is the most affected by the loss of observations as only transitions from one type of school to another permits identification of the private schooling effect. Comparing Samples 1 and 3, we observe that reducing our sample by excluding the first three cycles of NLSCY slightly reduces the private school effect. The reduction of the pri-

¹² As stated in the NLSCY's user guide for the first survey (1994–1995): "This scale was administered to the Person Most Knowledgeable (PMK) of the child, generally the child's mother, or to the spouse/partner on the Parent Questionnaire, and measures how family members relate to each other." More precisely: "Questions related to family functioning were developed by researchers at the Chedoke-McMaster Hospital of McMaster University and have been used widely both in Canada and abroad. This scale is used to measure various aspects of family functioning, e.g. problem solving, communications, roles, affective involvement, affective responsiveness and behavior control".

Table 9
Estimated effects of private school attendance on children's math percentile rank score, 7–15-year-old children from Cycles 1–7, and Cycles 4–7 of the NLSY.

Model	OLS		Fixed effects		Random effects	
Sample 1: all public and private schools children—Cycles 1–7						
Private school	9.87 (1.69)***	10.00 (1.66)***	5.39 (1.58)***	5.24 (1.70)***	7.66 (1.36)***	7.56 (1.41)***
Hyperactivity	–	–1.23 (0.17)***	–	0.09 (0.19)	–	–0.70 (0.16)***
Anxiety	–	0.38 (0.20)*	–	0.29 (0.21)	–	0.28 (0.18)
Aggression	–	–0.80 (0.31)**	–	–0.55 (0.33)*	–	–0.77 (0.28)***
N (children)	5156(1700)	4594(1691)	5156(1700)	4594(1691)	5156(1700)	4594(1691)
Sample 2: only end of primary to first secondary grades school children—Cycles 1–7						
Private School	10.31 (2.23)***	10.18 (2.30)***	7.28 (2.13)***	6.61 (2.51)***	8.65 (1.90)***	8.36 (2.04)***
Hyperactivity	–	–1.35 (0.28)***	–	–0.28 (0.38)	–	–1.11 (0.35)***
Anxiety	–	0.29 (0.30)	–	0.44 (0.37)	–	0.27 (0.29)
Aggression	–	–1.64 (0.47)***	–	–0.30 (0.64)	–	–1.48 (0.46)**
N (children)	2348(1,153)	1998(1123)	2348(1153)	1998(1123)	2348(1153)	1998(1123)
Sample 3: sample 1: all public and private schools children—Cycles 4–7						
Private School	9.83 (2.33)***	9.87 (2.33)***	4.71 (1.97)**	4.81 (2.11)**	6.77 (1.91)***	6.72 (2.00)***
Hyperactivity	–	–1.71 (0.23)***	–	–0.30 (0.25)	–	–1.11 (0.21)***
Anxiety	–	0.40 (0.24)	–	0.27 (0.26)	–	0.27 (0.22)
Aggression	–	–0.58 (0.38)	–	–0.04 (0.38)	–	–0.38 (0.33)
N (children)	3447(1176)	3067(1170)	3447(1176)	3067(1170)	3447(1176)	3067(1170)
Sample 4: Only end of primary to first secondary grades school children—Cycles 4–7						
Private School	9.85 (2.52)***	9.36 (2.59)***	3.95 (2.33)*	2.12 (2.73)	6.84 (2.11)***	6.34 (2.26)***
Hyperactivity	–	–1.73 (0.34)***	–	–0.69 (0.46)	–	–1.48 (0.32)***
Anxiety	–	0.31 (0.35)	–	0.27 (0.43)	–	0.24 (0.33)
Aggression	–	–1.22 (0.56)**	–	–0.17 (0.69)	–	–1.09 (0.53)**
N (children)	1751(864)	1502(840)	1751(864)	1502(840)	1751(864)	1502(840)

Notes: (1) Standard errors clustered on the individuals (children) in parentheses. Statistical significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. (2) Each behavioral score is centered on its mean and divided by its standard deviation for each grade level and cycle.

vate schooling effect is larger for Sample 4, particularly for the fixed effects estimate which dives down from 7.28 and 6.61 (with behavioral controls) to 3.95 to 2.12 (a joint null test of the behavioral controls is not rejected in the fixed effects model). We must remember that Sample 4 has approximately half the transitions observed in Sample 2, making identification more tenuous. Hence, attrition may play a role in the results, but the overall evidence that private schooling has a positive effect on mathematical skills remains quite strong.

Because the percentile rank dependent variable is truncated at 0 and 100, we estimated the model using as dependent variable the number of standard deviations from the mean (for children taking the test at the same grade). The results are presented in Table 10.

The estimated (positive) effects range from approximately 1 tenths of a standard deviation to almost 4 tenths of a standard deviation. The exclusion of Cycles 1–3 observations reduces the size of the effects as with percentile rank. The last panel shows that fixed effects estimates are no longer significant with behavioral scores, but these results are based on a very small number of transitions from private to public school, so that identification is tenuous.

There remains the problem of Ashenfelter's Dip. To address this issue, we compute for a sample composed of those who move from public to private school and the year they move, the first and second lag of the standardized score. We then take the difference between these two scores to observe whether there is a negative trend in scores inciting the parents to move their child to private school. Fig. 1 shows the distribution of these first-differences. They show no clear trend as the mean and median are close to 0 and the distribution very symmetric. This provides further evidence that parents are not reacting to drops in the score by sending their child to private school.

Our results show quite convincingly that private schools can considerably improve the achievement level of young teenagers. Although, we cannot affirm that randomly choosing a child in a public school and moving him to private school will improve his lot, we can affirm, given the credibility of our identifying assumptions, that parents who choose private schools are making a choice that, on average, will improve their children's test scores and this, at a relatively low cost for the family.

8. Interpretation

Why are the private schools producing better rankings and higher scores for the children of parents who choose them? Several factors could be in play.

One possible first answer is peer effects. As students move from public to private school, the average skill level increases substantially in their class because of the selection of students based on ability. A study by Boucher et al. (2010) using a very large 75% random sample of fourth and fifth grade secondary students from Québec public schools and the individual scores of students from four standardized tests taken in June 2005 (Math, Sciences, French and History), find statistically significant and positive endogenous peer effects for several topics (in the sense of Manski,

1993) and they are particularly high in Math. This study separates rigorously endogenous effects, from contextual or correlated effects using a method found in Bramoullé, Habiba, and Fortin (2009).

Although their results generally indicate that students benefit from their peers' higher test scores, they do not find significant effects of the contextual socio-economic mix of students. Comparing peer effects in private and public schools across countries, Zimmer and Toma (2000) do not find robust peer effects across school type or socio-economic mix (measured by parent's education). But they find larger significant peer effects for lower-ability students across school types. Mixing students by ability in schools appears to benefit students with lower ability. This is likely the quest of most families choosing a private school for their child without knowing in advance his or her ability. Most private schools select students on the basis of their entrance test. However, private schools will not admit only high ability students and not others, or all wealthy students and no others, but have to make a trade-off between those who can pay and those who increase the quality of the school (Walton, 2010). This may explain why most private schools offer partial scholarship each year for some students out of their endowment revenues.

The remaining answers are more hypothetical and are mostly anecdotal in nature. Private schools monitor more closely the performance of children immediately noticing parents of a failing grade and requiring that the child receive pedagogical support in the subject with poor grades. Thirdly, delinquent behavior in many cases will lead to immediate expulsion. Fourth, excellence is highly valued within schools and students are invited to participate as much as they can in scientific and artistic contests. Discipline is praised and valued. All students being accepted must sign the conduct code (wear uniforms or conform to the dress and hair codes), and politeness and courtesy are required at all times in the school possibly developing non-cognitive skills. The academic work load is more considerable if a student is to obtain the passing grade in private schools.

It is hard to measure the quality of teachers, but the working environment is superior in private schools because teachers do not have to deal with problem students and good teaching is rewarded by observing a large number of students quickly grow into outstanding students. Almost all teachers in the private sector are unionized in the same two unions representing all public teachers (in a distinct federation). Teaching loads and salaries are similar to the ones in the public system, being paid according to experience and diploma (credential). The low intensity of behavioral problems, students with on average higher ability, and an environment rewarding achievements and schooling progression may attract better teachers. Furthermore, they are not afraid of a policy ending public subsidies as they would find teaching posts in the public sector.

The answer certainly does not lie in the curriculum which is similar as all 4th and 5th secondary grade students in the province must pass the Ministry of Education exams to graduate from high school and subsidies are obtained on the condition that schools follow the same curriculum as public schools.

Table 10
Estimated effects of private school attendance on children's math standard deviation score, 7–15-year-old children from Cycles 1–7, and Cycles 4–7 of the NLSY.

Model	OLS		Fixed effects		Random effects	
Sample 1: all public and private schools children—Cycles 1–7						
Private school	0.35 (0.06)***	0.36 (0.06)***	0.24 (0.05)***	0.22 (0.06)***	0.29 (0.05)***	0.28 (0.05)***
Hyperactivity	–	–0.04 (0.01)***	–	0.00 (0.01)	–	–0.02 (0.01)***
Anxiety	–	0.01 (0.01)**	–	0.01 (0.01)	–	0.01 (0.01)*
Aggression	–	–0.02 (0.01)**	–	–0.02 (0.01)	–	–0.02 (0.01)**
N (children)	5156 (1700)	4594 (1691)	5156 (1700)	4594 (1691)	5156 (1700)	4594 (1691)
Sample 2: only end of primary to first secondary grades transitions for private schools children—Cycles 1–7						
Private school	0.36 (0.08)***	0.35 (0.08)***	0.28 (0.07)***	0.24 (0.08)***	0.31 (0.07)***	0.29 (0.07)***
Hyperactivity	–	–0.05 (0.01)***	–	–0.01 (0.01)	–	–0.04 (0.01)***
Anxiety	–	0.01 (0.01)	–	0.01 (0.01)	–	0.01 (0.01)
Aggression	–	–0.05 (0.02)***	–	0.00 (0.02)	–	–0.05 (0.02)***
N (children)	2948 (1153)	1998 (1123)	2348 (1153)	1998 (1123)	2348 (1153)	1998 (1123)
Sample 3: all public and private schools children—Cycles 4–7						
Private school	0.35 (0.07)***	0.35 (0.07)***	0.21 (0.06)***	0.20 (0.06)***	0.27 (0.05)***	0.26 (0.05)***
Hyperactivity	–	–0.06 (0.01)***	–	–0.01 (0.01)	–	–0.03 (0.01)***
Anxiety	–	0.02 (0.01)*	–	0.01 (0.01)	–	0.01 (0.01)
Aggression	–	–0.02 (0.01)*	–	–0.01 (0.01)	–	–0.02 (0.01)
N (children)	3695 (1246)	3299 (1241)	3695 (1246)	3299 (1241)	3695 (1246)	3299 (1241)
Sample 4: only end of primary to first secondary grades transitions for private schools children—Cycles 4–7						
Private school	0.36 (0.09)***	0.34 (0.09)***	0.18 (0.08)**	0.12 (0.09)	0.27 (0.07)***	0.24 (0.08)***
Hyperactivity	–	–0.06 (0.01)***	–	–0.03 (0.02)**	–	–0.05 (0.01)***
Anxiety	–	0.01 (0.01)	–	0.01 (0.01)	–	0.01 (0.01)
Aggression	–	–0.04 (0.02)**	–	–0.00 (0.02)	–	–0.04 (0.02)**
N (children)	1751 (864)	1502 (840)	1751 (864)	1502 (840)	1751 (864)	1502 (840)

Notes: (1) Standard errors clustered on the individuals (child) in parentheses. Statistical significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. (2) Each behavioral score is centered on its mean and divided by its standard deviation for each grade level and cycle.

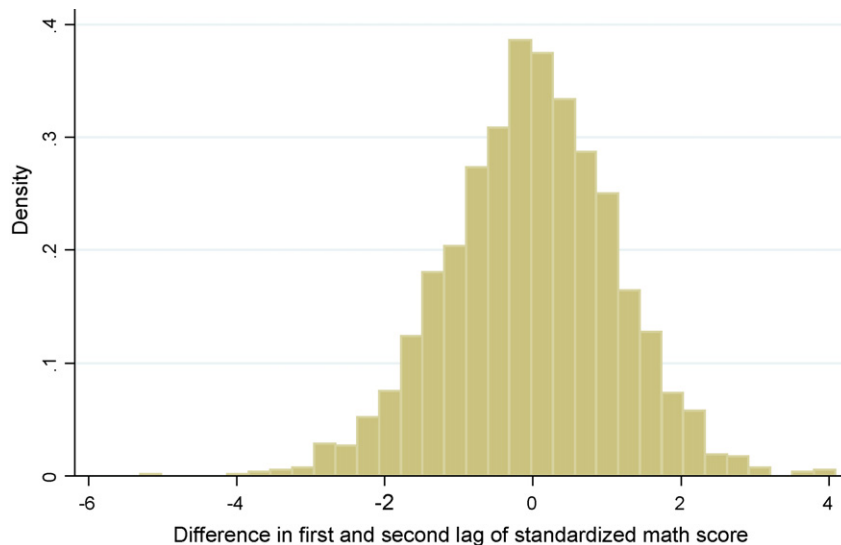


Fig. 1. Distribution of first difference in lagged standardized math scores switchers from elementary public school to private secondary school.

What about the external validity of the results? One aspect of Québec's dynamics that is particularly different from other provinces that have recently instituted a subsidized system for private schools is the history of the schooling system in Québec and the tradition of "elite" Catholic schools as of "elite" universities in the United States. There is very little public outcry towards the concept of public funding of private schools despite repeated attempts by teachers' unions, through media campaigns, to stop the subsidies. Anecdotal evidence points towards a tradition within families of private schooling, from fathers to sons and mothers to daughters. This tradition is certainly not as strong in other provinces where historical aspects do not play a key part, we believe, in parental schooling choices. The very long history of these schools is certainly advantageous as successful academic practices can be transmitted from one generation of principals and teachers to another. This would be the main reason, in our view, that the success of the Québec system may not be transported elsewhere in Canada and the United States. Table 6 shows that the distribution of students in private school between primary and secondary is more balanced than in Québec where a large majority is in secondary school, possibly showing structural differences between the Québec system and those in the Rest of Canada. It is difficult to explain why, but we know that in Québec historical factors strongly increased the supply of private secondary spaces. Despite these reservations on the external validity of the results, our paper still provides evidence that type of schooling can matter considerably for young teenage students.

9. Conclusion

This paper provides evidence that sending a child to a private school can, *ceteris paribus*, improve his/her ranking in mathematics. More precisely, it empirically supports the hypothesis that children who do attend private schools

in the province of Québec benefit from their parents' decision to send them to this type of school (a positive effect of the treatment on the treated). The effect is large as it ranges from 4 to 10 percentile ranks and 12% to 36% of a standard deviation. The size of these effects, although for only one outcome, a score on a math test, are similar to the effects estimated for American Catholic schools.

The evidence demonstrates the potential gains obtained from a competitive system of schools. For school year 2007–2008, private high schools educate 18% of high school students and 6.6% of primary school children in Québec. Table 7 shows that public schools have reacted to the flight of students to private schools by offering particular pedagogical projects mainly in urban settings.

These improvements are also achieved at a low cost for the provincial government. For school year 2004–2005, at primary and secondary schools, the subsidies to the private sector are \$416 million to which is added \$457 million from fees and other revenues (see Table 3), for total revenues of \$874 million. For the same year and school levels, the public sector received \$6131 millions of public subsidies, raised \$1233 million in school taxes, and obtained \$757 million in other revenues for total revenues of \$8120 million. The public and private sectors respectively schooled almost a million students and a little more than a hundred thousand students.

Our results provide some evidence that subsidizing private schools could be a wise policy to improve performance. Because we use relative measures of performance, it is difficult to ascertain whether such a policy would increase average test scores (e.g. of future PISA test score results), but given that children rank better once they move to a private school could demonstrate that they are better prepared for tests, more motivated and possibly be putting more effort in academics. We do obtain smaller effects when we control for attrition, but this is done by considerably reducing the sample size.

One of the main mechanisms that can justify these subsidies is the creation of a more competitive system with more choices (lifting partially and exerting pressures on the residential location constraints on public school enrollment) for parents that can raise the quality of all schools. In fact, some evidence against the “cream skimming hypothesis” and for the “tide lifting all boats” hypothesis are the results presented in [Table A4](#). When comparing the math scores for children at the same level of schooling in the public schools, we find that the students in Québec are on average doing better than those in the Rest of Canada despite the fact that a much larger proportion of Québec children are in private schools and are doing better than children in public schools. The sample of children in the Rest of Canada is much more representative of all children than the sample of children in public schools in Québec where the high-ability students are underrepresented. Therefore, competition could be driving up scores in public schools. The difference between children in Québec and the Rest of Canada is certainly not attributable to differences in income as the mean income of families with children in Québec is lower than for the Rest of Canada.

What do these results suggest for policy purposes? The Québec experience shows that opening up public schools to more competition; and some limited subsidies to private schools as well as a regulated environment for these schools is a prudent way to raise performance within and

across schools. To provide more evidence-based precise suggestions for policy, we need to answer directly the question of why private schools are more effective than public schools.

Acknowledgements

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Appendix A.

See [Tables A1–A4](#).

Table A1

Mean characteristics of children (7–15-year-olds), their PMK and family by cycle, overall sample (all children).

Characteristics	Cycle 1: 1994–1995	Cycle 2: 1996–1997	Cycle 3: 1998–1999	Cycle 4: 2000–2001	Cycle 5: 2002–2003	Cycle 6: 2004–2005	Cycle 7: 2006–2007
Sample	242	531	514	1,050	1,184	949	686
Child							
Age in months	115 ± 16	129 ± 22	137 ± 31	126 ± 28	132 ± 26	151 ± 19	167 ± 13
Private	0.02 [4]	0.07 [35]	0.07 [38]	0.06 [58]	0.08 [100]	0.12 [117]	0.17 [114]
Behavioral scores							
Hyperactivity	5.02 ± 3.94	4.09 ± 3.40	3.94 ± 3.33	3.63 ± 2.97	3.95 ± 2.93	3.61 ± 2.76	3.49 ± 2.7
Anxiety	3.19 ± 2.92	3.20 ± 2.64	3.00 ± 2.60	2.93 ± 2.49	3.17 ± 2.52	3.07 ± 2.55	2.93 ± 2.6
Aggression	1.38 ± 1.85	0.97 ± 1.67	0.97 ± 1.88	1.06 ± 1.75	1.14 ± 1.87	0.77 ± 1.45	0.68 ± 1.4
Repeated her last grade	0.09 [21]	0.05 [28]	0.05 [24]	0.06 [58]	0.06 [67]	0.04 [40]	0.12 [79]
Changed school	0.22 [54]	0.13 [67]	0.13 [65]	0.14 [149]	0.14 [161]	0.12 [110]	0.10 [69]
Is a girl	0.54 [130]	0.55 [292]	0.53 [271]	0.50 [523]	0.50 [592]	0.49 [464]	0.50 [342]
Has a chronic condition	0.22 [54]	0.25 [132]	0.26 [135]	0.31 [329]	0.29 [341]	0.28 [267]	0.29 [196]
PMK							
Birth age 18–25 years old	0.23 [55]	0.22 [119]	0.24 [122]	0.21 [223]	0.22 [260]	0.21 [203]	0.22 [150]
Birth age 26–30 years old	0.48 [115]	0.45 [237]	0.43 [221]	0.43 [449]	0.43 [504]	0.42 [394]	0.40 [276]
Birth age 31–35 years old	0.25 [60]	0.27 [145]	0.27 [138]	0.29 [305]	0.28 [336]	0.29 [278]	0.30 [203]
Birth age 36–40 years old	0.05 [12]	0.06 [30]	0.06 [33]	0.07 [73]	0.07 [84]	0.08 [74]	0.08 [57]
Primary education	0.21 [51]	0.20 [108]	0.17 [87]	0.13 [140]	0.13 [157]	0.13 [123]	0.12 [84]
Secondary diploma	0.24 [57]	0.21 [112]	0.18 [93]	0.20 [206]	0.20 [239]	0.20 [193]	0.21 [146]
Post-secondary	0.27 [66]	0.22 [118]	0.24 [123]	0.23 [243]	0.25 [292]	0.23 [222]	0.24 [163]
University diploma	0.28 [68]	0.36 [193]	0.41 [211]	0.44 [461]	0.42 [496]	0.43 [411]	0.43 [293]
Non-immigrant	0.97 [235]	0.97 [516]	0.97 [497]	0.97 [1,020]	0.97 [1,144]	0.97 [916]	0.96 [661]
Family							
Family functioning score	7.55 ± 5.49	7.63 ± 5.16	7.82 ± 4.59	8.16 ± 4.97	7.41 ± 4.93	7.31 ± 4.92	7.24 ± 4.9
One older child	0.36 [88]	0.37 [195]	0.32 [167]	0.33 [351]	0.35 [414]	0.35 [331]	0.36 [244]
At least 2 older children	0.17 [41]	0.14 [76]	0.10 [51]	0.11 [112]	0.10 [118]	0.09 [90]	0.10 [69]
At least 1 younger child	0.50 [121]	0.47 [247]	0.50 [259]	0.52 [550]	0.53 [628]	0.54 [510]	0.53 [364]
Biological/adoptive parents	0.80 [194]	0.79 [422]	0.81 [418]	0.77 [811]	0.75 [894]	0.73 [692]	0.69 [469]
Single parent	0.14 [34]	0.15 [77]	0.15 [75]	0.15 [159]	0.17 [202]	0.18 [173]	0.20 [140]
Step parent	0.06 [14]	0.06 [32]	0.04 [21]	0.08 [80]	0.08 [88]	0.09 [84]	0.11 [73]
Not a big city (<500,000)	0.82 [198]	0.74 [393]	0.74 [378]	0.71 [741]	0.71 [843]	0.72 [681]	0.66 [451]
Family total income	53,683 ± 29,433	55,539 ± 33,642	65,001 ± 44,083	66,949 ± 43,330	66,489 ± 39,942	67,671 ± 39,979	71,551 ± 42,673

Notes: (1) The interpretation of each behavioral score is similar: the higher the score, the more the child is afflicted by the problem the score represents. (2) Percentage of observations and number of observations in brackets. (3) Family income in \$2002 Canadian.

Table A2

Mean characteristics of children (7–15-year-olds), their PMK and their family by cycle, restricted sample.

Characteristics	Cycle 1: 1994–1995	Cycle 2: 1996–1997	Cycle 3: 1998–1999	Cycle 4: 2000–2001	Cycle 5: 2002–2003	Cycle 6: 2004–2005	Cycle 7: 2006–2007
Sample	86	190	217	318	478	661	398
Child							
Age in months	133 ± 6	145 ± 13	145 ± 15	144 ± 14	145 ± 16	148 ± 15	161 ± 8
Private	0.00 [0]	0.05 [10]	0.05 [11]	0.05 [17]	0.09 [45]	0.08 [51]	0.08 [51]
Behavioral scores							
Hyperactivity	4.51 ± 3.93	3.71 ± 3.01	3.53 ± 3.20	3.27 ± 2.58	3.56 ± 2.52	3.62 ± 2.67	3.48 ± 2.5
Anxiety	3.24 ± 2.88	3.44 ± 2.89	3.22 ± 2.57	3.15 ± 2.48	2.99 ± 2.39	3.01 ± 2.49	2.94 ± 2.5
Aggression	1.16 ± 1.55	0.91 ± 1.70	0.89 ± 1.71	0.82 ± 1.55	1.00 ± 1.80	0.75 ± 1.45	0.71 ± 1.4
Repeated her last grade	X [<5]	0.05 [9]	0.04 [9]	0.06 [18]	0.07 [32]	0.04 [24]	0.12 [46]
Changed school	0.21 [18]	0.11 [21]	0.11 [24]	0.13 [41]	0.13 [62]	0.11 [74]	0.11 [44]
Is a girl	0.56 [48]	0.56 [106]	0.55 [119]	0.51 [161]	0.45 [217]	0.49 [322]	0.52 [206]
Has a chronic condition	0.20 [17]	0.25 [48]	0.25 [55]	0.31 [99]	0.29 [138]	0.27 [179]	0.28 [112]
PMK							
Birth age 18–25 years old	0.23 [20]	0.23 [44]	0.24 [52]	0.22 [70]	0.20 [97]	0.21 [140]	0.23 [93]
Birth age 26–30 years old	0.40 [34]	0.40 [76]	0.42 [92]	0.44 [139]	0.44 [211]	0.43 [283]	0.41 [163]
Birth age 31–35 years old	0.30 [26]	0.29 [55]	0.28 [61]	0.30 [95]	0.28 [136]	0.28 [186]	0.29 [113]
Birth age 36–40 years old	0.07 [6]	0.08 [15]	0.06 [12]	0.04 [14]	0.07 [34]	0.08 [52]	0.07 [28]
Primary education	0.19 [16]	0.21 [39]	0.19 [41]	0.13 [41]	0.13 [61]	0.13 [84]	0.13 [52]
Secondary diploma	0.27 [23]	0.21 [39]	0.19 [41]	0.20 [64]	0.19 [93]	0.21 [140]	0.23 [91]
Post-secondary	0.27 [23]	0.25 [48]	0.25 [55]	0.24 [75]	0.25 [120]	0.24 [159]	0.23 [90]
University diploma	0.28 [24]	0.34 [64]	0.37 [80]	0.43 [138]	0.43 [204]	0.42 [278]	0.41 [165]
Non-immigrant	1.00 [86]	0.99 [189]	0.99 [214]	0.98 [312]	0.98 [468]	0.97 [642]	0.97 [385]
Family							
Family functioning score	7.31 ± 5.37	8.02 ± 5.42	8.01 ± 4.30	8.12 ± 4.85	6.99 ± 4.83	7.28 ± 4.95	7.01 ± 4.9
One older child	0.36 [31]	0.35 [66]	0.34 [74]	0.35 [110]	0.33 [158]	0.36 [237]	0.38 [153]
At least 2 older children	0.17 [15]	0.15 [28]	0.12 [26]	0.11 [34]	0.08 [38]	0.10 [65]	0.12 [46]
At least 1 younger child	0.49 [42]	0.49 [94]	0.43 [93]	0.53 [167]	0.59 [282]	0.54 [355]	0.49 [197]
Biological/adoptive	0.73 [43]	0.80 [152]	0.83 [179]	0.78 [249]	0.74 [354]	0.73 [483]	0.68 [271]
Single parent	0.20 [17]	0.15 [29]	0.14 [31]	0.14 [43]	0.19 [89]	0.18 [120]	0.20 [80]
Step parent	0.07 [6]	0.05 [9]	0.03 [7]	0.08 [26]	0.07 [35]	0.09 [58]	0.12 [47]
Not a big city (<\$500,000)	0.78 [67]	0.75 [143]	0.76 [165]	0.73 [232]	0.75 [357]	0.73 [481]	0.65 [260]
Family total income	51,289 ± 27,939	53,799 ± 30,262	65,262 ± 45,334	70,296 ± 47,765	66,621 ± 38,415	66,021 ± 35,235	70,075 ± 36,008

Notes: (1) The interpretation of each behavioral score is similar: the higher the score, the more the child is afflicted by the problem the score represents. (2) Percentage of observations and number of observations in brackets. (3) Family income in \$2002 Canadian. (4) When a cell has fewer than 5 observations, Statistics Canada's rule does not permit that the information be presented.

Table A3

Mean math standardized test score by age and cycle.

Age of children	Mean score (standard deviation) [number of observations]									
	Cycle 1: 1994–1995	Cycle 2: 1996–1997	Cycle 3: 1998–1999	Cycle 4: 2000–2001	Cycle 5: 2002–2003	Cycle 6: 2004–2005	Cycle 7: 2006–2007			
7-year-olds	324 (46) [35]	336 (49) [39]	312 (35) [55]	301 (40) [175]	–	–	–			
8-year-olds	365 (48) [59]	396 (47) [64]	368 (48) [88]	339 (51) [170]	346 (41) [205]	–	–			
9-year-olds	429 (53) [47]	448 (56) [88]	395 (41) [13]	399 (60) [166]	386 (46) [246]	–	–			
10-year-olds	496 (47) [53]	481 (48) [93]	474 (51) [59]	437 (54) [168]	427 (50) [175]	416 (54) [222]	–			
11-year-olds	531 (55) [48]	514 (56) [83]	498 (47) [61]	476 (55) [101]	468 (56) [168]	439 (49) [238]	–			
12-year-olds	–	563 (60) [88]	527 (62) [83]	513 (52) [88]	500 (56) [141]	491 (67) [150]	476 (60) [210]			
13-year-olds	–	608 (77) [76]	535 (58) [49]	552 (73) [83]	533 (69) [95]	520 (70) [141]	502 (68) [206]			
14-year-olds	–	–	640 (100) [56]	593 (102) [62]	575 (86) [80]	570 (88) [115]	560 (97) [143]			
15-year-olds	–	–	669 (101) [50]	607 (92) [50]	631 (91) [74]	612 (91) [83]	597 (102) [127]			

Authors' calculation from the transversal micro-data sets of the NLSY. The test is a shorter version of the Mathematics Computation Test-CAT/2.

Note: Collection for the first cycle of the NLSY began in 1994 with one large cohort of 0–11-year-olds. At Cycle 6 (7) there are no children from age 7 to 9 (10 and 11).

Table A4

Average standardized math scores of high school students in public schools by grade in Québec and in the other provinces.

Secondary grade	Québec	Other provinces
7	504	473
8	534	511
9	589	571
10	634	599
All	548	528

Authors' calculation from the weighted transversal micro-data sets of the NLSY, Cycles 1–7.

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